Linear Regression

- Lecture 4 · Introduction to Machine Learning

Outline

- Supervised learning recap
- Linear models
- Least squares
- Gradient descent
- Regularisation
- Evaluation

Supervised learning

- Given pairs (x, y), learn a function f so that f(x) ≈ y
- Regression: y is a real number
- Classification: y is a label

Training data

- Dataset $D = \{(x_i, y_i)\}$ for $i = 1..n$
- Features $x_i \in R^d$
- Targets $y_i \in R$

The linear model

- Prediction $\hat{y} = w^{\top}x + b$
- w: weights, b: bias (intercept)
- Linear in the parameters, not necessarily in x

Feature maps

- Replace x with $\varphi(x)$ to fit curves
- Polynomial features: $\varphi(x) = (1, x, x^2, x^3)$
- Still a linear model in w

Loss functions

- Measure how wrong predictions are
- Squared error: $(y - \hat{y})^2$
- Absolute error: $|y - \hat{y}|$

Least squares objective

- Minimise $J(w) = \Sigma\ (y_i - w^T x_i)^2$
- Convex in w → single global minimum
- Matrix form: $J(w) = ||y - Xw||^2$

Normal equations

- Set gradient to zero: $X^T(Xw - y) = 0$
- Solution: $w^* = (X^TX)^{-1} X^Ty$
- Requires X^TX to be invertible

Geometry of least squares

- Xw^* is the projection of y onto the column space of X
- Residual $y - Xw^*$ is orthogonal to the columns of X

Cost of the closed form

- Inverting X^TX costs $O(d^3)$
- Slow when d is large
- Motivates iterative methods

Gradient descent

- Start from w_0
- Repeat: $w \leftarrow w - \eta \nabla J(w)$
- η is the learning rate

Choosing the learning rate

- Too small: slow convergence
- Too large: divergence or oscillation
- Use a schedule or line search

Stochastic gradient descent

- Use one example (or a mini-batch) per step
- Noisy but cheap updates
- Scales to very large datasets

Feature scaling

- Standardise each feature to mean 0, variance 1
- Makes gradient descent converge faster

Overfitting

- Model fits noise in the training data
- Low training error, high test error
- More likely with many features

Ridge regression

- Add L2 penalty $\lambda||w||^2$
- Closed form: $w^* = (X^TX + \lambda I)^{-1} X^Ty$
- Shrinks weights towards zero

Lasso regression

- Add L1 penalty $\lambda||w||_1$
- Produces sparse weights
- No closed form; use coordinate descent

Choosing λ

- Hold out a validation set
- k-fold cross-validation
- Pick λ with the lowest validation error

Bias–variance trade-off

- Bias: error from wrong assumptions
- Variance: error from sensitivity to data
- Regularisation trades variance for bias

Evaluation metrics

- Mean squared error (MSE)
- Root mean squared error (RMSE)
- Coefficient of determination R^2

Interpreting R^2

- $R^2 = 1 - SS_{res} / SS_{tot}$
- 1 means perfect fit, 0 means no better than the mean

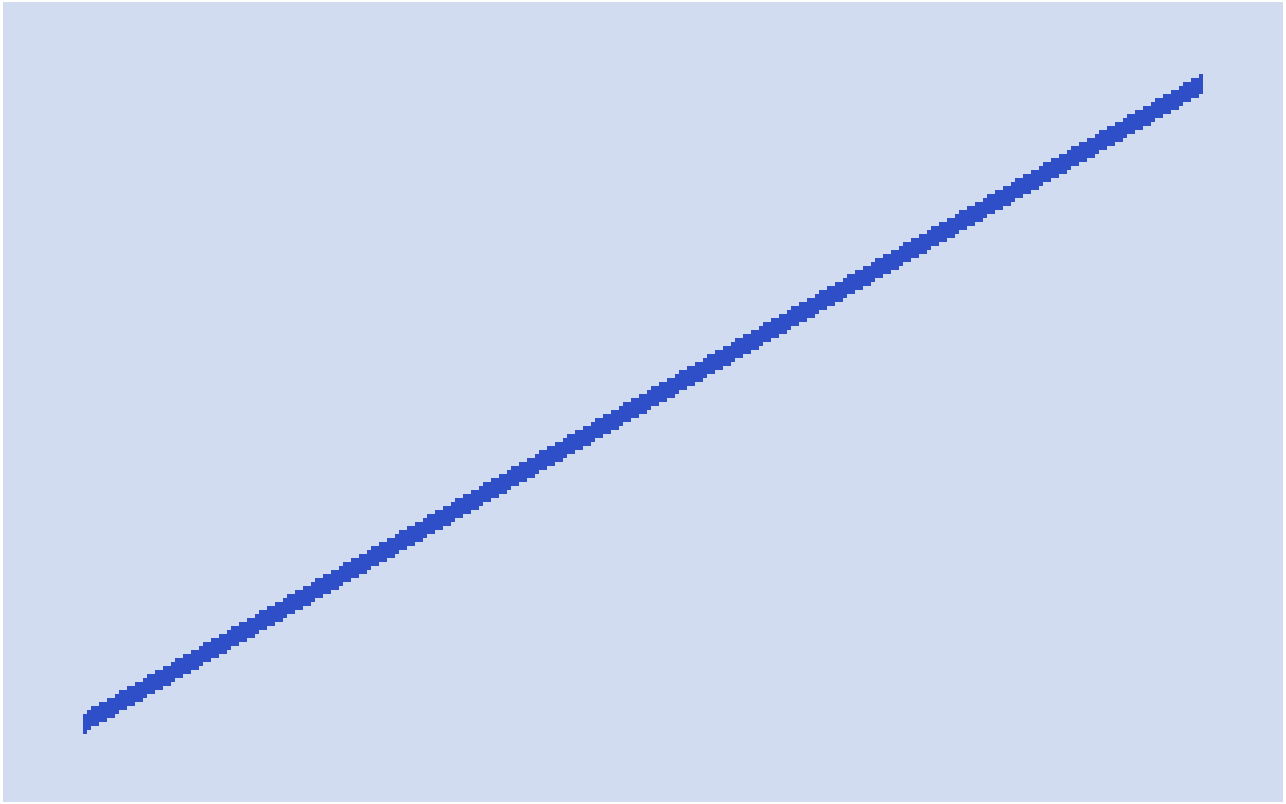


Fig. 3

Summary

- Linear models predict with $w^T x + b$
- Least squares has a closed form
- Gradient descent scales better
- Regularise to avoid overfitting

Questions?